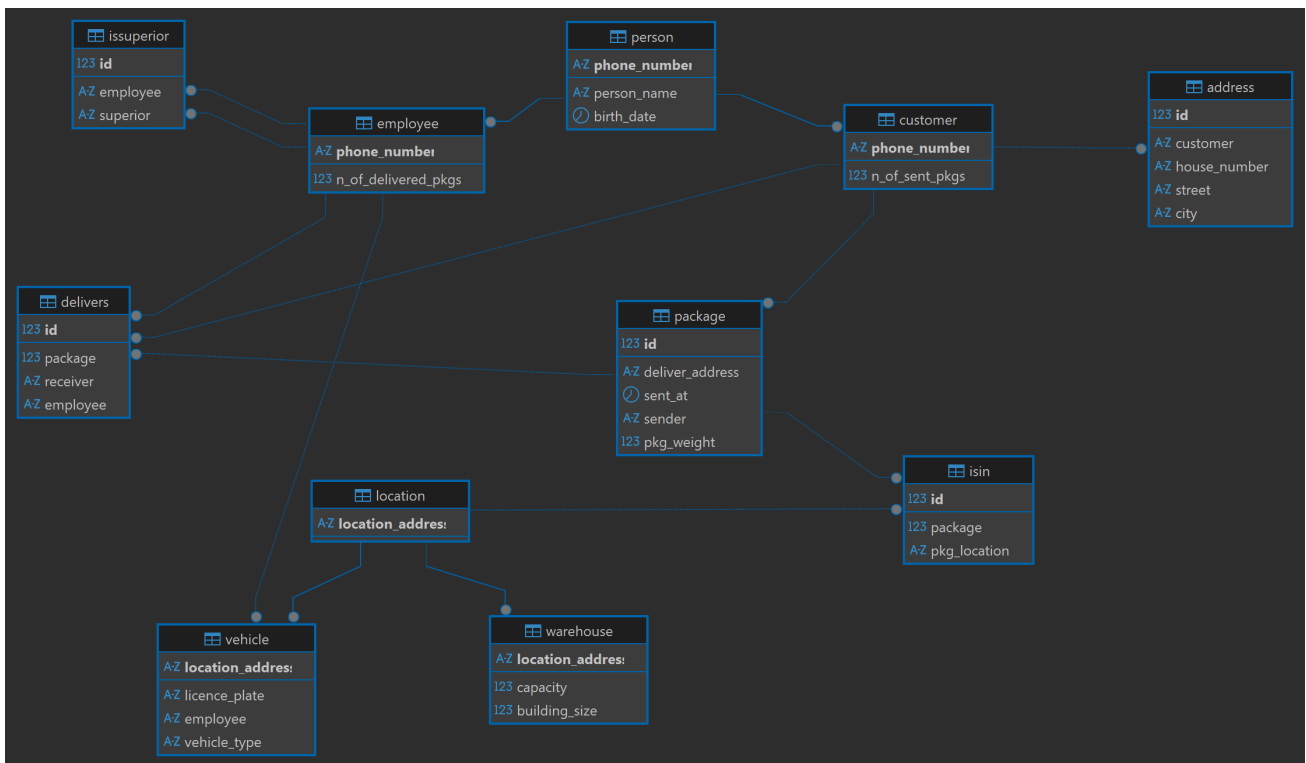


CP3

Model databáze

- Person(name, birth_date, phone_number)
- Customer(phone_number, address)
 - FK: phone_number ⊆ Person(phone_number)
- Address(phone_number, house_number, street, city)
 - FK: phone_number ⊆ Customer(phone_number)
- Employee(phone_number, #_of_delivered_packages)
 - FK: phone_number ⊆ Person(phone_number)
- IsSuperior(employee, superior)
 - FK: employee ⊆ Employee(phone_number)
 - FK: superior ⊆ Employee(phone_number)
- Location(address)
- Vehicle(address, licence_plate, employee, type)
 - FK: address ⊆ Location(address)
 - FK: employee ⊆ Employee(phone_number)
- Warehouse(address, capacity, building_size)
 - FK: address ⊆ Location(address)
- Package(deliver_address, timestamp, sender, weight)
 - FK: sender ⊆ Customer(phone_number)
- IsIn(pkg_deliver_address, pkg_timestamp, pkg_sender, location)
 - FK: (pkg_deliver_address, pkg_timestamp, pkg_sender) ⊆ Package(deliver_address, timestamp, sender)
 - FK: location ⊆ Location(address)
- Delivers(pkg_deliver_address, pkg_timestamp, pkg_sender, receiver, employee)
 - FK: (pkg_deliver_address, pkg_timestamp, pkg_sender) ⊆ Package(deliver_address, timestamp, sender)
 - FK: receiver ⊆ Customer(phone_number)
 - FK: employee ⊆ Employee(phone_number)



SQL dotazy na vytvoření tabulek

```
CREATE TABLE Person (  
    phone_number char(9) primary key check (phone_number ~ '^[0-9]{9}$'),  
    person_name varchar(100) not null,  
    birth_date date not null  
)  
  
CREATE TABLE Customer (  
    phone_number char(9) primary key references Person(phone_number) on update cascade on delete cascade,  
    n_of_sent_pkgs int not null default 0 check (n_of_sent_pkgs >= 0)  
)  
  
CREATE TABLE Address (  
    id int generated always as identity primary key,  
    customer char(9) not null references Customer(phone_number) on update cascade on delete cascade,  
    house_number varchar(100) not null,  
    street varchar(100) not null,  
    city varchar(100) not null,  
    unique(customer, house_number, street, city)  
)  
  
CREATE TABLE Employee (  
    phone_number char(9) primary key references Person(phone_number) on update cascade on delete cascade,  
    n_of_delivered_pkgs int not null default 0 check (n_of_delivered_pkgs >= 0)  
)  
  
CREATE TABLE IsSuperior (  
    id int generated always as identity primary key,  
    employee char(9) not null references Employee(phone_number) on update cascade on delete cascade,  
    superior char(9) not null references Employee(phone_number) on update cascade on delete cascade,  
    unique(employee, superior)  
)  
  
CREATE TABLE Location (  
    location_address varchar(300) primary key  
)  
  
CREATE TABLE Vehicle (  
    location_address varchar(300) primary key references Location(location_address) on update cascade on delete restrict,  
    licence_plate char(7) not null unique,  
    employee char(9) references Employee(phone_number) on update cascade on delete set null,  
    vehicle_type varchar(100) not null  
)  
  
CREATE TABLE Warehouse (  
    location_address varchar(300) primary key references Location(location_address) on update cascade on delete restrict,  
    capacity int not null,  
    building_size int not null check (building_size >= 0)  
)  
  
CREATE TABLE Package (  
    id int generated always as identity primary key,  
    deliver_address varchar(300) not null,  
    sent_at timestamp not null,  
    sender char(9) not null references Customer(phone_number) on update cascade on delete restrict,  
    pkg_weight int not null check (pkg_weight >= 0),  
    unique(deliver_address, sent_at, sender)  
)  
  
CREATE TABLE IsIn (  
    id int generated always as identity primary key,  
    package int not null references Package(id) on update cascade,  
    pkg_location varchar(300) not null references Location(location_address) on update cascade on delete restrict,  
    unique(package, pkg_location)  
)  
  
CREATE TABLE Delivers (  
    id int generated always as identity primary key,  
    package int not null references Package(id) on update restrict,  
    receiver char(9) not null references Customer(phone_number) on update cascade on delete restrict,  
    employee char(9) not null references Employee(phone_number) on update cascade on delete restrict,  
    unique(package, receiver, employee)  
)
```

SQL selecty

INNER JOIN

```
select c.phone_number, a.city from customer c join address a on c.phone_number = a.customer order by c.phone_number
```

Vypisuje tel. číslo zákazníka a město, ve kterém bydlí, seřazené podle tel. čísla.

	AZ phone_number	AZ city
1	700000001	Brno
2	700000002	Ostrava
3	700000003	Plzeň
4	700000004	Liberec
5	700000005	Olomouc
6	700000006	Pardubice
7	700000007	Hradec Králové
8	700000008	Zlín
9	700000009	České Budějovice
10	700000010	Praha
11	700000011	Brno
12	700000012	Ostrava

OUTER JOIN

```
select p.id as package_id, d.employee as employee_phone from package p left join delivers d on d.package = p.id
```

Vypisuje balíčky a v případě, že je někdo doručuje, tak i tel. číslo doručovatele (jinak NULL).

	123 package_id	AZ employee_phone
1	5,927	[NULL]
2	30,212	[NULL]
3	10,448	[NULL]
4	11,474	[NULL]
5	11,920	[NULL]
6	11,782	[NULL]
7	15,386	[NULL]
8	20,802	[NULL]
9	3,345	[NULL]
10	10,942	[NULL]
11	3,811	[NULL]
12	14,278	[NULL]

PODMÍNKA NA DATA

```
select p.phone_number, p.person_name, c.n_of_sent_pkgs from customer c join person p on c.phone_number = p.phone_number where c.n_of_sent_pkgs > 100
```

Vypisuje informace uživatelů, kteří odeslali nejméně 100 balíčků.

	AZ phone_number	AZ person_name	123 n_of_sent_pkgs
1	700000001	Petr B. Svoboda	921
2	700000002	Josef C. Novotný	936
3	700000003	Pavel D. Dvořák	904
4	700000004	Martin E. Černý	882
5	700000005	Tomáš F. Procházka	915
6	700000006	Lukáš G. Kučera	938
7	700000007	Jakub H. Veselý	959
8	700000008	Marek I. Horák	946
9	700000009	Jiří J. Němec	903
10	700000010	Karel K. Marek	939
11	700000011	Michal L. Pokorný	605
12	700000012	David M. Pospíšil	578

VNOŘENÝ SELECT

```
select w.location_address, w.capacity from warehouse w where w.capacity > (select avg(w2.capacity) from warehouse w2)
```

Vypisuje sklady, jejichž kapacita je větší než průměrná kapacita skladu.

	AZ location_address	123 capacity
1	Lokace 151, 251, Brno	2,010
2	Lokace 152, 252, Ostrava	2,020
3	Lokace 153, 253, Plzeň	2,030
4	Lokace 154, 254, Liberec	2,040
5	Lokace 155, 255, Olomouc	2,050
6	Lokace 156, 256, Pardubice	2,060
7	Lokace 157, 257, Zlín	2,070
8	Lokace 158, 258, Jihlava	2,080
9	Lokace 159, 259, Kladno	2,090
10	Lokace 160, 260, Praha	2,100
11	Lokace 161, 261, Brno	2,110
12	Lokace 162, 262, Ostrava	2,120

ŘAZENÍ A STRÁNKOVÁNÍ

```
select p.id, p.sent_at from package p order by p.sent_at desc limit 10 offset 10
```

Vypisuje druhých deset balíčků seřazených sestupně podle data odeslání.

	123 id	sent_at
1	30,527	2025-03-26 05:01:37.000
2	9,377	2025-03-26 04:54:07.000
3	11,177	2025-03-26 04:24:07.000
4	12,977	2025-03-26 03:54:07.000
5	14,777	2025-03-26 03:24:07.000
6	16,577	2025-03-26 02:54:07.000
7	18,377	2025-03-26 02:24:07.000
8	20,177	2025-03-26 01:54:07.000
9	21,977	2025-03-26 01:24:07.000
10	827	2025-03-26 01:16:37.000

PODMÍNKA NA AGREGAČNÍ FUNKCI

```
select d.employee, count(*) as delivered_count from delivers d group by d.employee having count(*) < 250 order by count(*) desc limit 10
```

Vypisuje deset nejlepších zaměstnanců, kteří doručili méně než 250 balíčků.

	AZ employee	123 delivered_count
1	700000148	249
2	700000137	246
3	700000142	245
4	700000149	243
5	700000135	243
6	700000129	239
7	700000145	235
8	700000150	234
9	700000147	231
10	700000139	229

MNOŽINOVÉ OPERACE

`select p.sender, p.id as package_id from package p where id in (select id from package except select package from delivers)`

Vypisuje tel. číslo zákazníka a id balíčku, kde balíček nebyl zatím doručen.

	A-Z sender	123 package_id
1	700000006	31,789
2	700000003	20,549
3	700000019	23,000
4	700000059	23,154
5	700000047	2,204
6	700000021	15,805
7	700000005	23,986
8	700000039	25,110
9	700000003	25,574
10	700000054	12,251
11	700000002	11,158
12	700000016	22,029